AQUAREV WATER · COMMERCIAL

Specifications & Installation Instructions

The written companion to Drawing Set 100: a three-part specification for the AquaRev water conditioning device on a commercial pool or spa recirculation system, and the step-by-step instructions the installing contractor works from.

Revision A

Issued for review · 22 September 2026

Companion to

Installation Drawing Set 100

Product Specification Guide 2026

Pump Room Implementation Guide Rev 1.1

aquarevwater.us/data

What this document is for

Part A is the specification: what is being installed, to which standards, and how the work is accepted. It is written in the three-part format engineers, facilities departments and authorities expect, so it can be attached to a purchase order, a contractor scope, or a permit submittal without rewriting. Part B is the installation instruction the crew carries into the pump room.

A **Specification · Section 22 51 19**
Swimming pool water treatment equipment: hydrodynamic cavitation water conditioning device

A.1 **Part 1 · General**
Summary, references, submittals, quality, handling, warranty

A.2 **Part 2 · Products**
Manufacturer, device, models, supplied and contractor materials

A.3 **Part 3 · Execution**
Examination, preparation, installation, field quality control, closeout

B **Installation instructions**
Before install day, the eight steps, after start-up, troubleshooting

C **Drawing cross-reference**
Which sheet answers which question



Installation Drawing Set 100 · Rev A

Eight ANSI B sheets: cover and notes, site plan, pump room plan, flow diagram, connection details, installation sequence, alternate arrangements, schedules and records. Every sheet reference in this document (for example 3/M-301) points into that set.

OPEN THE DRAWING SET | DOWNLOAD PDF, 17 × 11

THE ONE RULE THAT NEVER CHANGES

The AquaRev device supplements the pool's disinfection system. It does not replace it. Free chlorine and pH must still be held inside the regulated range, tested on schedule, and recorded, exactly as before.

PART A · SPECIFICATION

Section 22 51 19 · Swimming Pool Water Treatment Equipment

Hydrodynamic cavitation water conditioning device, in-line, passive. Section numbering follows CSI MasterFormat so the section drops into a project manual beside the existing pool plumbing sections. Where a project uses a different numbering, keep the clause text and renumber.

PART 1 General

1.1 SUMMARY

1.1.A **Work included.** Furnish and install one AquaRev water conditioning device per recirculating loop, in series on the existing return line, downstream of all treatment equipment, using the two Schedule 40 PVC unions supplied with the device, and with the records listed in Part 3.

1.1.B **Work not included.** No modification to the pump, filter, heater, chlorinator or salt cell, chemical feeders, controller, sensors, suction piping, main drains, drain covers, shell, deck or barriers. No electrical, gas, water-supply, drain, sewer or backwash connection. No change to turnover, flow rate or disinfection method.

1.1.C **Related documents.** Installation Drawing Set 100, sheets G-001 through M-501. Product Specification Guide 2026. Pump Room Implementation Guide Rev 1.1 (site assessment, approvals, permit determination, jurisdiction package). Installation Pathway Rev 1.0 (who does what, when).

1.2 REFERENCES

REFERENCE	APPLIES TO
NSF/ANSI/CAN 50-2024	Equipment for swimming pools, spas, hot tubs. The device is certified to this standard by IAPMO R&T.
IAPMO R&T Product Listing, File No. W-12547	Listing of the device as a Water Conditioning Device bearing the USPC® mark, listed in compliance with the Uniform Swimming Pool, Spa and Hot Tub Code. Verify at pld.iapmo.org.
NSF/ANSI 372	Lead content. Device materials are lead-free.

REFERENCE	APPLIES TO
ASTM D1785	PVC pipe, Schedule 40, for any reroute or replacement lengths.
ASTM D2466	PVC Schedule 40 socket fittings: couplings, elbows, tees.
ASTM F656	Primer for PVC solvent-cemented joints.
ASTM D2564	Solvent cement for PVC pipe and fittings.
ASTM D2855	Standard practice for the two-step (primer and solvent cement) method of joining PVC pipe and fittings.
Local codes	Building or plumbing code and public pool health code as adopted by the authority having jurisdiction, and that authority's written permit determination.

1.3 DEFINITIONS

1.3.A **Device, WCD.** The AquaRev hydrodynamic cavitation water conditioning device, including its two union halves and O-rings.

1.3.B **Loop.** One recirculating system: one body of water with its own pump, filter and return. Each loop receives its own device.

1.3.C **L1, L2, D1, D2.** Overall length including unions; body length collar to collar; body diameter; union nut diameter. Per Product Specification Guide 2026 and sheet M-301.

1.3.D **Cut length.** The length of existing pipe removed, measured socket bottom to socket bottom with both union tails dry-fitted to the device. It is always less than L1.

1.3.E **AHJ.** Authority having jurisdiction: the building department and the public-pool health program, or whichever office decides on changes to public pool equipment.

1.4 SYSTEM DESCRIPTION AND PERFORMANCE

1.4.A The device is a passive PVC pressure vessel with fixed internal geometry. Water flowing through it undergoes hydrodynamic cavitation. It has no moving parts, no media, no consumables, no electrical or control connection, and requires no calibration.

1.4.B Rated head loss is 2.6 psi at maximum rated flow for every model. Flow ranges by model are in 2.3. The loop's design flow must fall inside the selected model's range.

1.4.C The device operates in any orientation. Gravity has no effect on performance.

1.4.D The device supplements disinfection. Sanitizer and pH requirements of the health code are unchanged and remain the operator's responsibility.

1.5 SUBMITTALS

1.5.A **Before ordering:** site assessment record for each loop (Implementation Guide §3) stating return pipe outside diameter and material, measured flow, filter gauge with clean filter, straight run available, isolation and drain-down method, and selected model.

1.5.B **Product data:** Product Specification Guide 2026 pages for the selected model; IAPMO R&T Product Listing W-12547, both pages; warranty terms as issued.

1.5.C **Before install day:** the AHJ determination record with the authority's written reply, or the permit; installer's licence and insurance per the property's vendor requirements; confirmed cut length from the unit in hand.

1.5.D **Closeout:** completed installation record (sheet M-501) with serial number, cut length, cure time, gauge readings before and after, leak checks at start-up and after 24 hours, and photographs of the carton label, the cut zone with a tape in frame, and the finished installation with the listing mark visible.

1.6 QUALITY ASSURANCE

1.6.A Device shall bear the IAPMO R&T listing mark. Substitutions are not permitted.

1.6.B Installer shall be a licensed pool contractor or plumber experienced in solvent-cemented PVC pressure piping. The warranty requires professional installation.

1.6.C Solvent-cemented joints shall follow ASTM D2855 and the cement manufacturer's instructions for pipe size, temperature and cure. Where these instructions and this document differ, the manufacturer's instructions govern.

1.6.D Refer to AquaRev before ordering when loop flow is within 10% of a model's range limit, the pool exceeds 500,000 gallons, or the return pipe is 8 in. or larger.

1.7 DELIVERY, STORAGE AND HANDLING

1.7.A Deliver in the manufacturer's carton. Carton sizes: 36 × 10.5 × 10.5 in. for 2, 3 and 4 in. models (8, 20 and 35 lb); 44 × 16.5 × 16.5 in. for the 6 in. model (50 lb).

1.7.B On receipt, photograph the carton label and confirm the model against the assessment record. Open and confirm both union halves and both O-rings are present. Store indoors, off the floor, out of direct sun, with the carton closed until install day.

1.8 PROJECT CONDITIONS

1.8.A Pool closure of three to five hours per loop, including cement cure. Schedule with operations; post signage and barrier per property policy.

1.8.B Ventilate the equipment room while priming and cementing: open door or fan. No heat or open flame.

1.8.C Cure time depends on pipe size and temperature. Cool or damp rooms need longer. Plan the closure window around the cement manufacturer's table, not around the pool schedule.

1.9 WARRANTY

1.9.A Lifetime limited warranty on the device body. Unions, O-rings and external piping are excluded. Conditions: professional installation following AquaRev's documentation, and water kept balanced within industry ranges. Claims to water@aquarevwater.us with install date and serial number. Full terms at aquarevwater.us/warranty.

PART 2 Products

2.1 MANUFACTURER

2.1.A AquaRev Water, aquarevwater.us. Sales and AHJ package support: water@aquarevwater.us, (832) 979-6758. Technical support and warranty: aquarev@hdcproductsinc.com, (808) 628-0208.

The IAPMO R&T listing carries the brand name HDC Products and the listee KM LLC dba KD Enterprises, Honolulu, HI. Cite the listing exactly as issued in any submittal.

2.2 DEVICE

2.2.A **Type:** Water Conditioning Device for pools, spas and hot tubs, certified to NSF/ANSI/CAN 50-2024 by IAPMO R&T, File No. W-12547.

2.2.B **Materials:** PVC body and fittings. NSF/ANSI 372 lead-free. California Proposition 65 compliant. Made in the USA. No metal components; no grounding required.

2.2.C **Connections:** Two Schedule 40 PVC unions, supplied, one each end, sized to the return pipe. Solvent-weld socket on the pipe side; O-ring face seal; hand-tightened nut.

2.2.D **Marking:** Label with listing mark, model and serial number, and IN and OUT flow direction.

2.3 MODELS

RETURN PIPE	2 IN.	3 IN.	4 IN.	6 IN.
Product ID	KD2X4X2	KD3X6X3	KD4X8X4	KD6X12X6
Flow range	44–127 gpm	112–273 gpm	240–480 gpm	440–1,100 gpm
Head loss, rated	2.6 psi	2.6 psi	2.6 psi	2.6 psi
Fitting	2 in. PVC hub	3 in. PVC hub	4 in. PVC hub	6 in. PVC hub
L1 overall	20.25 in.	24.00 in.	28.19 in.	40.00 in.
L2 body	13.375 in.	15.00 in.	18.88 in.	25.00 in.
D1 body dia.	5.00 in.	7.25 in.	9.34 in.	13.75 in.
D2 union dia.	3.60 in.	5.06 in.	6.19 in.	9.00 in.
Min. straight run, L1 + 8 in.	28.25 in.	32.00 in.	36.19 in.	48.00 in.
Clear around body, D1 + 4 in.	9.00 in.	11.25 in.	13.34 in.	17.75 in.
Shipping weight	8 lb	20 lb	35 lb	50 lb

Eight-inch and ten-inch units are available on request. Source: Product Specification Guide 2026. Measure the unit in hand before marking pipe; dimensions are for selection and clearance, not for cutting.

2.4 SUPPLIED WITH THE DEVICE

2.4.A Two Schedule 40 PVC union halves with O-rings, matched to the device size. Nothing else is required from AquaRev for a standard installation.

2.5 MATERIALS FURNISHED BY THE CONTRACTOR

ITEM	REQUIREMENT
Primer	ASTM F656, purple, within shelf date.
Solvent cement	ASTM D2564 for PVC. Heavy-body cement for 4 in. and larger. Within shelf date.
Pipe	ASTM D1785 Schedule 40 PVC, same size as the return line. Carry 10 ft for reroute or recovery.
Fittings	ASTM D2466 Schedule 40 PVC socket couplings, 90° elbows and tees as required for arrangements C and D on sheet M-401.
Pipe supports	None required for the device. Where a rerouted run (arrangement C on M-401) needs support, corrosion-resistant strut or clevis hangers with clamps.
Lock-out	Lock-out devices and tags for the pump disconnect and every chemical feeder on the loop.
Label	Optional flow-direction label on the pipe at the device: arrow toward pool, "AquaRev WCD, remove at unions".

PART 3 Execution

3.1 EXAMINATION

3.1.A Walk the equipment room with a tape and camera and complete the assessment record for each loop. Confirm: return pipe outside diameter and material; flow from the flow meter or pump curve; filter gauge with a clean filter; the last treatment component on the return; straight run available after it; isolation valves or drain-down method; injection points and sample taps; door and access for the carton.

3.1.B Confirm the selected model's flow range brackets the measured flow, and that pump margin covers an added 2.6 psi. Where the straight run is less than L1 + 8 in., plan the offset reroute (M-401, C). Where flow exceeds a model's maximum, plan parallel devices (M-401, D) with AquaRev's concurrence.

3.1.C Report unforeseen conditions before cutting: non-PVC return pipe, pipe outside the union's tolerance, no way to isolate or drain, or a return run that cannot be made to fit.

3.2 PREPARATION

3.2.A Approvals in hand: spend approval, brand engineering acceptance where applicable, the AHJ's written permit determination or permit, and the scheduled closure window. The contractor carries a copy of the determination on site.

3.2.B Materials on site per 2.5. Device unpacked, label photographed, serial recorded, union halves and O-rings counted, IN and OUT identified.

3.2.C Dry-fit both union tails to the device and measure socket bottom to socket bottom. Record the cut length before anything is switched off.

3.3 INSTALLATION

3.3.A **Lock-out.** Switch off and lock the pump at its disconnect and every chemical feeder on the loop, including controller dosing outputs, before opening the line. Depressurize; confirm the filter gauge reads zero.

3.3.B **Location.** On the return line, downstream of the pump, filter, heater, chemical injection points and sensor taps, as close to the pool return as practical. Nothing is installed downstream of the device except existing pipe. Existing isolation valves, if any, remain in place.

3.3.C **Orientation.** Any. IN toward the pump. Label facing out where it can be photographed.

3.3.D **Cut.** Square, at the marked cut length, with 4 in. minimum clear from each union to any fitting. Deburr inside and out; wipe clean and dry.

3.3.E **Joints.** Slide the nut onto the pipe first. Prime pipe end and socket, cement both, push fully home with a quarter turn, hold 30 seconds, wipe the bead. Repeat the other end. Orient nuts to be turned by hand. Follow ASTM D2855 and the cement manufacturer.

3.3.F **Set.** Seat the device in the tails with O-rings in place and faces clean. Confirm full insertion. Tighten each nut hand-tight plus a small snug. No wrench, no thread sealant, no cement on the threads.

3.3.G **Support.** No dedicated supports are required; the device is carried by the return line on its two unions. Existing pipe supports remain in place. On a long unsupported horizontal run, support the pipe or the device body adjacent to the device as good practice.

3.3.H **Cure.** Full cure per the cement manufacturer for the pipe size and temperature before pressurizing. Close drains and refill the filter during the wait.

3.4 FIELD QUALITY CONTROL

3.4.A Remove the pump lock-out only. Start with the air relief open until water runs clear. Inspect both unions and both welds under operating pressure. A weep at a union is corrected by depressurizing, loosening, cleaning the faces, reseating the O-ring and retightening by hand, never by more torque.

3.4.B Read the filter gauge. Accept an increase of not more than 2.6 psi over the clean-filter reading before installation. Record both.

3.4.C Restore feeders and controller last. Recheck for leaks after 24 hours and record.

3.5 START-UP AND FIRST WEEKS

3.5.A Controller setpoints are not changed. Dosing continues as the health code requires; any reduced demand shows up as less dosing, not as a reason to run leaner.

3.5.B Expect faster filter loading in week one; a backwash or cartridge clean is normal. Chemistry stays inside the regulated range, which is also a warranty condition.

3.6 CLOSEOUT

3.6.A Project file contains: assessment record with photographs; AHJ determination record with the written reply or permit and inspection sign-off; listing, specification guide and warranty as issued; completed installation record with serial numbers, installer licence and photographs.

3.7 MAINTENANCE

3.7.A None scheduled. The device has no media, screens, lamps, cartridges or chemicals.

3.7.B If a blockage is suspected after the filter has been cleaned: lock out pump and feeders, loosen both unions, remove the device, turn it 180°, run water backward through it for five minutes, and reinstall in the original direction. Then correct whatever let debris past the filter.

3.7.C Winterize with the rest of the loop. The device drains through the unions.

PART B · INSTALLATION INSTRUCTIONS

For the crew in the pump room

One loop, one device, one crew, three to five hours including cure. Almost all of the risk sits in three places: feeders left live, a cut made to the catalogue length instead of the measured one, and pressure applied before the cement has cured. The eight steps below match sheet M-302 panel for panel.

Before install day

- [ ] **Assessment record complete**
  Pipe OD and material, flow, clean-filter gauge, straight run, isolation, model.
- [ ] **AHJ determination in hand**
  Written reply or permit. Carry a copy on site.
- [ ] **Closure scheduled and posted**
  Three to five hours per loop.
- [ ] **Device received and checked**
  Label photographed, serial recorded, two union halves and two O-rings present.
- [ ] **Cut length measured from the unit**
  Tails dry-fitted, socket bottom to socket bottom. Written down.
- [ ] **Primer and cement in date**
  Heavy-body cement for 4 in. and up.
- [ ] **Spare pipe and fittings on the truck**
  Same size and schedule as the return line.
- [ ] **Lock-out kit for pump and every feeder**
  Locks, tags, and a way to ventilate the room.

> **LOCK-OUT FIRST, EVERY TIME**
> The pump disconnect and every chemical feeder on the loop: chlorinator or salt cell, acid or CO_2 feed, and the controller's dosing outputs. Stopping flow with feeders live can concentrate chemical in the line and off-gas when the pipe is opened.

The eight steps

01 Lock out and tag `M-302 · 01`

1. Switch off the pump at its disconnect. Lock and tag.
2. Switch off and lock every feeder on the loop and the controller's dosing outputs. Tag each.
3. Confirm nothing can restart: no timer, no remote, no second switch.

02 Isolate and depressurize `M-302 · 02`

1. Close return-line isolation valves where fitted. Otherwise open the filter drain and let the line empty.
2. Open the filter air relief. The gauge must read zero before any cut.
3. Bucket under the cut zone. Open the door or run a fan.

03 Measure, mark and cut `M-302 · 03 and 3/M-301`

1. Mark the recorded cut length on the return run, after the last treatment component and as close to the pool return as the room allows, with 4 in. clear each side to any fitting.
2. Measure again. A short cut is recovered with a coupling and spare pipe; a long cut is not.
3. Cut square at both marks. Remove the section.

Never cut to L1. L1 includes the union tails, which slide over the pipe you are keeping.

04 Debur and clean `M-302 · 04`

1. Remove burrs inside and out on both cut ends. Wipe ends and sockets clean and dry.
2. Dry-fit a tail on each end. It should enter one third to two thirds of the socket depth by hand. If it will not enter, or drops fully home loose, stop and check pipe size and schedule.

05 Prime and cement the union tails `M-302 · 05 and 2/M-301`

1. Slide the union nut onto the pipe first, facing the cut. It cannot be added later.
2. Prime the pipe end and the tail socket. Cement both while the primer is wet.
3. Push the tail fully home with a quarter turn. Hold 30 seconds. Wipe the bead.
4. Repeat on the other end. Set both nuts so they can be turned by hand once the device is in.

06 Set the device `M-302 · 06`

1. Check the IN arrow points toward the pump, label facing out.
2. O-rings in their grooves, faces clean and dry. Seat the device in the tails and confirm full insertion.
3. Tighten each nut hand-tight plus a small snug. No wrench. No thread sealant. No cement on the threads. The O-ring makes the seal, not torque.
4. Leave the existing pipe supports where they are. No dedicated support is needed for the device.

07 Cure, then refill M-302 · 07

1. Wait the cement manufacturer's full cure for the pipe size and room temperature. Two-inch joints at room temperature typically want one to two hours; 3 and 4 in. two to four; 6 in. four or more; cool or damp rooms longer. When in doubt, wait longer.
2. Meanwhile close drains, refill the filter, leave the air relief open, tidy the cut zone.

08 Restart, check and record M-302 · 08

1. Open isolation valves. Remove the pump lock-out only.
2. Start the pump with the air relief open until water runs clear. Close the relief.
3. Inspect both unions and both welds under pressure. A weep at a union: depressurize, loosen, clean faces, reseat the O-ring, retighten by hand.
4. Read the filter gauge. Expect up to 2.6 psi more than the clean-filter reading, not more.
5. Restore feeders and controller. Photograph the installation with the listing mark visible.
6. Fill the installation record on sheet M-501. Recheck for leaks after 24 hours.

If the straight run is too short

Use arrangement C on sheet M-401: remove the short run and rebuild it as an offset with four 90° elbows of the same size and schedule, putting the device on the new horizontal leg. Support the new run at its elbows as for any rerouted pipe; the device itself needs none. Four elbows on 4 in. pipe at 417 gpm add well under 1 psi; confirm pump margin on the record. The device stays the last component.

If the loop's flow is above the model's range

Use arrangement D on sheet M-401: identical parallel branches with isolation valves, each device seeing half the flow, inside its range. The single model that matches the return pipe is always the first choice; parallel branches are for flow the single model cannot take. Confirm with AquaRev before ordering.

Troubleshooting after start-up

SYMPTOM	DO THIS
Weep at a union	Depressurize, loosen, clean both faces, reseat the O-ring, retighten by hand. Do not add torque, sealant or cement.
Gauge up more than 2.6 psi	Clean the filter first; that is the cause far more often than the device. If still high with a clean filter, check the device for debris per 3.7.B.
Filter loading faster in week one	Normal. Agglomerated fines reach the filter for the first time. Backwash or clean as usual.
Controller dosing less	Expected. Setpoints stay where the health code puts them; the controller finds the new demand on its own.
Device suspected blocked	Lock out pump and feeders. Loosen unions, remove, turn 180°, run water backward 5 minutes, reinstall in original direction. Fix what let debris past the filter.
Winter	Winterize with the rest of the loop. The device drains through the unions.

Quick reference card

Print and carry

LOCATION	Return line. After pump, filter, heater, feeders, sensors. Last component. Any orientation. IN toward pump.
CUT LENGTH	Socket bottom to socket bottom, tails dry-fitted. Never L1. 2 in. field example: L1 20.25 in., cut 17.5 in.
CLEARANCE	Straight run L1 + 8 in. Body D1 + 4 in. Hand access to both nuts.
JOINTS	Nut on first. Primer F656, cement D2564, heavy body at 4 in. and up. Full depth, quarter turn, 30 s.
NUTS	Hand tight plus a small snug. No wrench, no dope, no cement on threads.
SUPPORT	None required. Existing supports remain. On a long unsupported run, support near the device.
CURE	Manufacturer's table for size and temperature. Longer when cold or damp.
RESTART	Pump lock-out off only. Air relief open till clear. Inspect. Gauge up to +2.6 psi. Feeders last. Photo. Record. Recheck 24 h.
HELP	Technical: aquarev@hdcproductsinc.com, (808) 628-0208. Sales and AHJ: water@aquarevwater.us, (832) 979-6758.

PART C · DRAWING CROSS-REFERENCE

Which sheet answers which question

QUESTION	SHEET	WHAT IT SHOWS
What are the rules and symbols?	G-001	General notes, abbreviations, legend, design basis, references.
Where is the work on the property?	C-101	Pool, spa, buried lines, equipment room, limit of work. All work is inside the room.
Where exactly does the device go?	M-101	Equipment plan at 1/2 in. = 1 ft. Both loops, cut zones, keynotes, dimensions to the device.
What is upstream and downstream?	M-201	Single-line flow diagram for pool and spa. Hydraulic check table.
How big is it and how does the joint work?	M-301	Device elevation with L1, L2, D1, D2; union joint section; cut-length detail; pipe support.
What is the order of work?	M-302	Eight steps from lock-out to restart, drawn.
The run is short, vertical, or the flow is too high?	M-401	Horizontal, vertical, offset reroute, parallel devices, clearance envelope by model.
What do I order and what do I hand in?	M-501	Device schedule, materials, tools, sequence check, installation record, closeout, contacts.

Open Drawing Set 100 · Download the PDF

TYPICAL DRAWINGS

Set 100 shows a typical arrangement: a 150,000-gallon pool and a 3,000-gallon spa sharing one equipment room. Before use on a property, replace the design basis with the site assessment record, fill the property and location fields in each title block, and have the set checked. Permit determinations rest with the authority having jurisdiction.